KnockKnock

Low-Cost IoT Intrusion Detection for Doors and Windows

IoT: Algorithms and Services

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Objectives

- Low-power intrusion detection for doors and windows
- Reduce energy via hardware (ESP32-C3, TLC393) and duty-cycle design
- Compare connectivity: ESP-NOW vs BLE
- Show sensor-to-hub pipeline with vibration detection and alerting

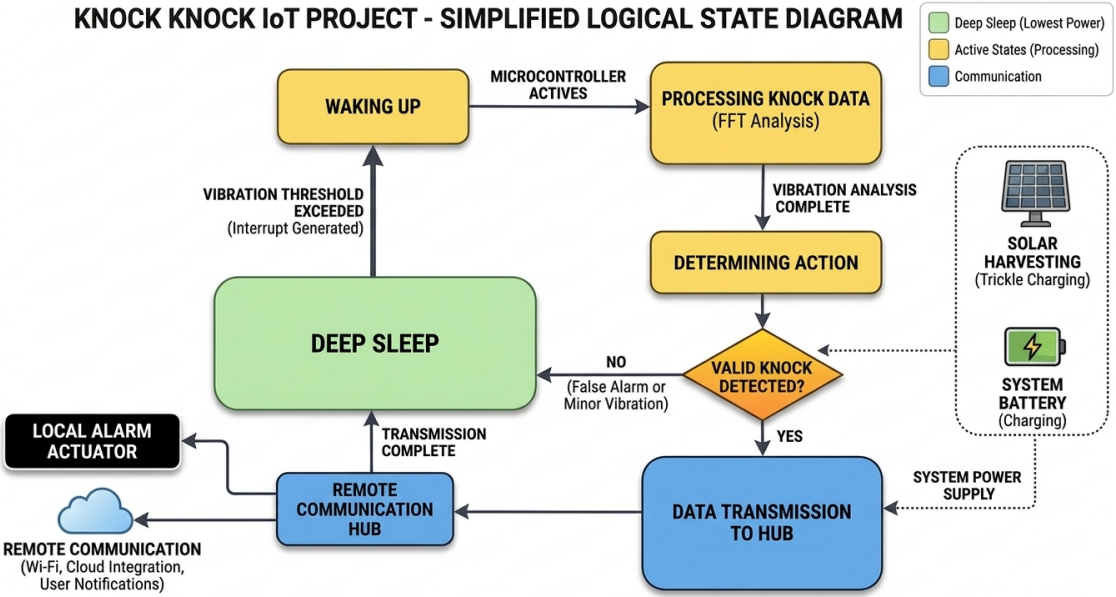
Architecture Overview

- **Two-tier** network: edge nodes + central hub
- Edge nodes: piezo sensor + LM393 + ESP32 V1
- Hub: always-on ESP32, **[TODO]** bridges to cloud via Wi-Fi
- Communication: **ESP-NOW** (node $\rightarrow$ hub)
- Cloud: **[TODO] MQTT** broker + web dashboard

Design Principle

Sense and classify at the edge, act at the hub.

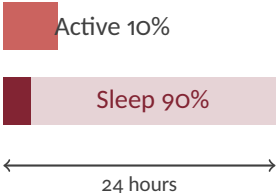
KNOCK KNOCK IoT PROJECT - SIMPLIFIED LOGICAL STATE DIAGRAM



Duty Cycle

2 Architecture

- **Alarm rate:** 12 alarms/hour = 288 alarms/day
- **Active time per alarm:** 30 s $\Rightarrow$ 86 400 s/day total wake time (worst case: 1 alarm/s)
- **Duty cycle:** 10%

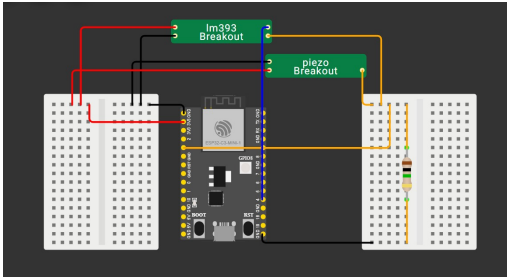


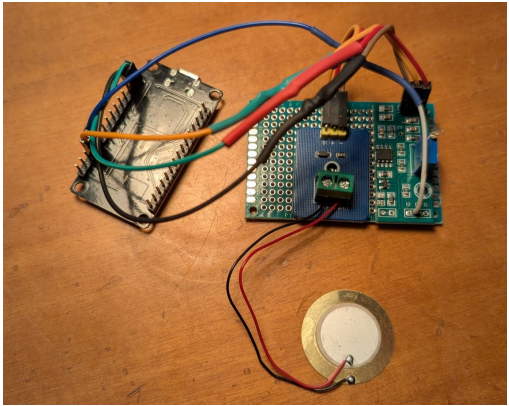
Key principle

The MCU is **off by default.**
The comparator never sleeps.

Wiring & Photo

2 Architecture





Wake-up: LM393 $\rightarrow$ TLC393

3 Improvements

Initial choice

- LM393 comparator
- 2.5 mA deep sleep

Result

- TLC393 comparator
- 11 μA deep sleep
- **227$\times$ less current**

$$2.5\,\text{mA} \xrightarrow{\text{swap}} 11\,\mu\text{A}$$

Protocol: BLE $\rightarrow$ ESP-NOW

3 Improvements

Initial choice

- BLE for local comms
- $\sim$ 500 ms pairing
- $\sim$ 7.5 ms TX latency

Result

- ESP-NOW
- **No pairing overhead**
- $<$ 2 ms TX

Shorter radio-on time $=$ **less energy per alarm**

Acquisition: Polling → Continuous DMA

3 Improvements

Initial choice

- Polling ADC reads
- CPU blocked 100%
- Missed samples at speed

Result

- DMA continuous mode
- **20 kHz**, CPU free

CPU free during capture = **FFT runs in parallel**

MCU: ESP32 V1 $\rightarrow$ ESP32-C3 RISC-V

3 Improvements

Initial choice

- ESP32 DOIT DevKit V1 dual-core
- Higher idle current: $\sim$4 mA (deep sleep)
- Oversized for the task

Result

- ESP32-C3
- 50 μA deep sleep
- 20 mA active

Right-sized MCU = **months of battery life**

Duty Cycle – Energy Calculation (Post-improvements)

Post-improvement parameters

- Deep sleep: MCU 50 μA + TLC393 11 μA $\rightarrow I_{sleep} = 61\,\mu$A
- ESP32-C3 active baseline: $I_{active} = 20$ mA
- Peak TX: 200 mA (1.5 ms per alarm), 288 alarms/day
- Duty cycle: $D = 10\%$
- Upper bound for wifi TX: 200 mA $\times$ 1 s

Average current

$$I_{avg} = D \cdot I_{active} + (1 - D) \cdot I_{sleep} = 0.10 \times 26\,\text{mA} + 0.90 \times 0.061\,\text{mA} \approx 2.65\,\text{mA}$$

Estimated battery life

$$t_{life} = \frac{2600\,\text{mAh}}{2.65\,\text{mA}} \approx 981\,\text{h} \approx 41\,\text{days}$$

Uses measured/spec'd post-improvement values; TX peaks are negligible on the daily average.

KnockKnock *Thank you for listening!*

Any questions?